SightOcc: Geometric Observability for Missed Objects in Camera-Only 3D Occupancy

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Abstract

Camera-only 3D occupancy models are evaluated as though every voxel were equally observable, yet a model can assign maximal confidence to free space exactly where the camera rig supplies almost no geometric evidence. Consequently, a missed object and correctly observed empty road are both simply free. *We introduce* SIGHTOCC*, a learning-free continuous measure of* geometric observability $O(v; \mathcal{G})$ *that combines projected voxel coverage with first-hit visibility across the rig. Unlike a binary camera-observation mask it carries evidence* strength *and resolves occlusion explicitly, and on Occ3D-nuScenes it ranks per-voxel error above the dataset's own* `mask_camera` *flag by +0.0484 AUROC on a sealed test split. Our main finding is an inversion across granularity. Model confidence is the stronger predictor of voxel-level error (0.8797 against 0.6321). For detecting objects that are missed entirely, geometric observability is the more informative signal (0.7623 against 0.7193, +0.0432 over 40,863 objects), and it comes within noise of a LiDAR point-count oracle. Confidence is a statement about a prediction that was made, and is structurally unable to flag one that was never made at all. Because observability is resolved against an external geometry field, we use it as an evaluation-time diagnostic rather than an inference-time signal.*

1. Introduction

A camera-only occupancy model does not abstain when queried about a volume outside its field of view. It predicts free space, and it does so confidently. Nothing in the model's output distinguishes "there is nothing here" from "I have no evidence either way", because both arrive as a high-probability vote for the free class. On the held-out scenes used throughout this paper, 47.3% of annotated dynamic objects standing in volumes with no camera evidence are not predicted by the model at all. Figure 1 shows one of them.

This matters because of what consumes the occupancy field. End-to-end planners interpret occupancy predictions as statements about the actual state of the world [9, 25] and are benchmarked on how well they act on it [11], and the functional-safety framing for exactly this hazard class (a system behaving as designed on input it was never equipped to interpret) has its own standard [28]. A planner reads the field as a statement about the world, not as a statement about the camera rig: a cell marked free is a cell the vehicle may drive through. The representation has no channel in which to say *I have no evidence here*; thus, absence of evidence is delivered to the planner as evidence of absence, and the planner has no way to tell the two apart. On the frames we measure, 76.6% of ground-truth obstacle voxels lie in volumes with no camera evidence at all.

The usual answer is to calibrate the model's own confidence. We show this answer is adequate for one question and structurally inadequate for another, and that the boundary between them is the granularity of the thing being asked about.

Our claim is sharper than "uncertainty estimation is hard": *a score computed from the network's output cannot by itself distinguish a confidently empty location from a confidently missed object.* A missed object is not a low-confidence prediction that the right machinery could surface. It is a confident prediction of the free class, which is indistinguishable, in the output alone, from correctly observed empty road. Calibration rescales confidences; it does not create a prediction that was never made. If the claim is right, the remedy cannot come from the model, and has to come from something that knows about the sensors.

Two different questions. Confidence asks whether the prediction the model just produced is likely to be correct. Observability asks whether the sensor configuration could have supplied evidence about that location at all. The two coincide when the model predicts an object and diverge when it confidently predicts free space, since there is no object hypothesis whose confidence can be evaluated. No recalibration of the output changes this: calibration rescales the confidences attached to predictions that exist. That is the

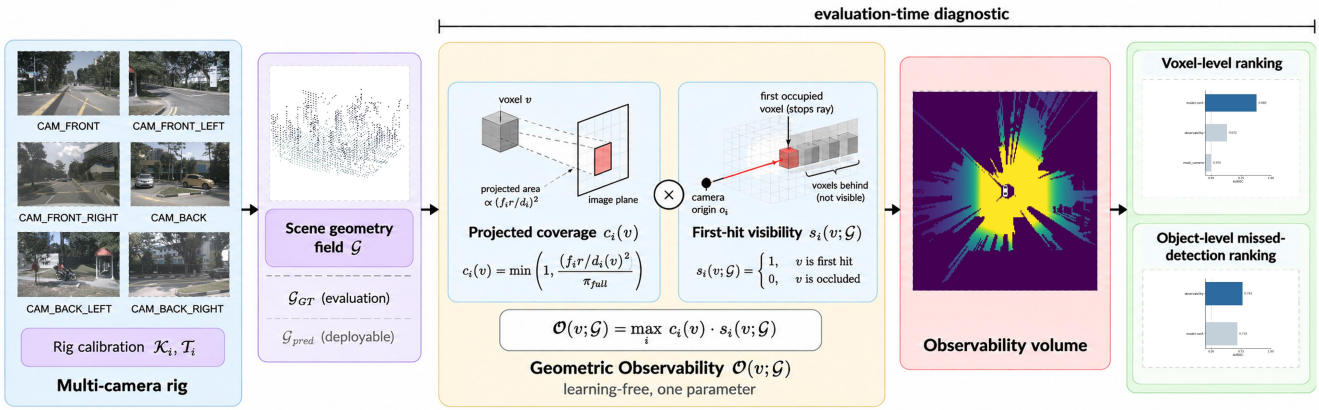


Figure 1. **SIGHTOCC: geometric observability from the camera rig alone.** The six rig images and the calibration define, for every voxel, a projected coverage $c_i(v)$, the image area the voxel subtends in camera i, and a first-hit visibility $s_i(v;\mathcal{G})$, which asks whether the ray from that camera reaches it through the scene-geometry field $\mathcal{G}$. Their product, maximised over cameras, is $O(v;\mathcal{G})$. No training and one parameter. Nothing in the pipeline reads the occupancy model, which is why the measure can rank failures the model's own confidence cannot: at object granularity observability reaches 0.7623 AUROC against 0.7193 for confidence, while at voxel granularity the ordering is the other way round. Rig images, geometry volume and observability map are one sealed-test frame, token `312cd4d3`; the two bar panels are the full test split.

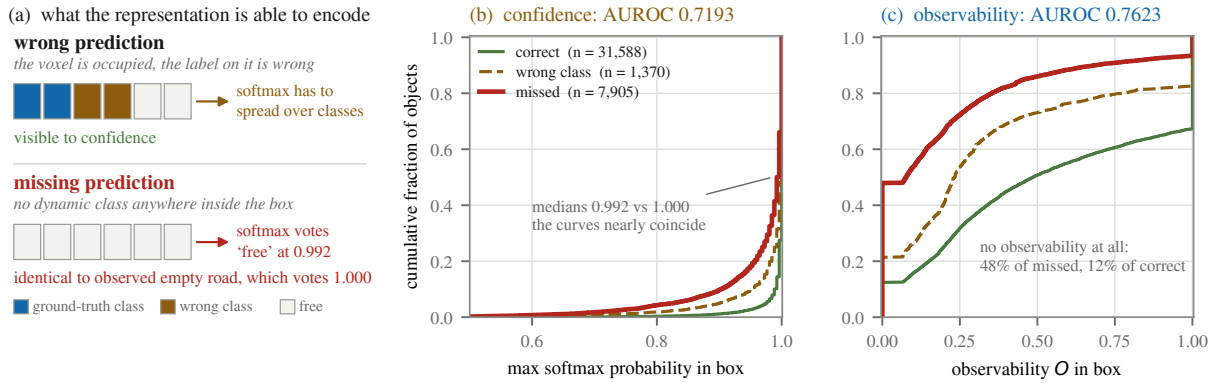


Figure 2. **Confidence can only rate a prediction that exists.** (a) The mechanism. A wrong prediction occupies the voxel with the wrong label and leaves a confidence trace there. A *missing* prediction occupies it with `free`, which is the same thing the model writes over correctly observed empty road, so there is no trace to read. (b, c) The same claim measured, on all 40,863 objects of the sealed test split, split by what the prediction placed inside the ground-truth box. Under maximum softmax the missed objects sit almost on top of the correctly detected ones (medians 0.992 and 1.000; 33.9% of missed objects carry a value of 1.000), which is why confidence reaches only 0.7193. Observability separates the same two populations (47.9% of missed objects against 12.4% of detected ones have no observability at all) and reaches 0.7623. Every quantity on this figure is written by `scripts/viz/fig_whyconf.py` to `fig_whyconf.json`.

gap external geometric evidence might fill, and the rest of this paper measures whether it does.

We therefore propose *geometric observability*, a measure computed outside the network from the camera rig and a scene-geometry field:

1. it is a function of the camera rig, the ego pose and the scene's occupancy, and of nothing the network produces, so it is unchanged when the model changes and transfers between architectures;
2. it is continuous, where the dataset's `mask_camera` flag is binary, so it is constant over the support where we compare and cannot rank within it; and
3. it carries exactly one parameter, fixed once at 965.0 and never retuned per model or per split.

Contributions.

- **Geometric observability.** We introduce SIGHTOCC, a continuous, learning-free, sensor-side measure that com-

bines projection coverage with first-hit visibility under an *explicit* scene-geometry field $\mathcal{G}$, carries one fixed parameter, and reads nothing from the occupancy model. The measure, the name and the protocol around it are introduced in this paper.

- **The granularity inversion.** Model confidence is the strongest signal for ranking voxel-level error. For objects the model fails to predict at all, observability is the more informative signal. Ensemble disagreement offers no advantage over confidence, while combining geometry with confidence outperforms either alone. The contrast is measured over 40,863 dynamic objects, and the two signals remain non-redundant at matched confidence.
- **An identified mechanism for the geometry source.** We show that what the object-level advantage depends on is the *density* of the occluder field, not a circularity in the idea: substituting the model's own dense occupancy overoccludes regions the rig genuinely sees, and we isolate which side of the comparison moves.
- **A registered evaluation protocol.** The experiments cover three camera-only models on Occ3D-nuScenes. We report every contrast with its paired scene-bootstrap confidence interval and its registration status, including the registrations that failed.

2. Related Work

Three literatures meet here. One supplies the model whose failures we study, one supplies the baseline we compare against, and one supplies the quantity we compute. None of them asks whether sensor geometry ranks failures better than the network's own confidence.

2.1. Camera-only occupancy prediction

Dense occupancy replaced boxes as the camera-only output of choice because it describes arbitrary geometry. The line runs from the lifting operation of Lift-Splat-Shoot [47] through the bird's-eye-view detectors built on it [8, 22, 26, 38] to BEVFormer's spatiotemporal attention [40]. Occupancy methods then split by how they treat the volume: as semantic scene completion [7, 39], as three orthogonal planes [27], as a dual-path transformer [59], as a direct multi-camera regression [55], or as forward-backward projection with a long temporal window [41]. Later work targets sparsity and cost [32, 43, 58, 62], fusion [36], and forecasting [37, 61]; a survey covers the area [60]. Occ3D [52] is the benchmark most are measured on, alongside OpenOccupancy [53] and SemanticKITTI [2], over nuScenes [6] and Waymo Open [50].

All of them report mIoU, an average over voxels. An average over voxels cannot say that some of those voxels were never visible.

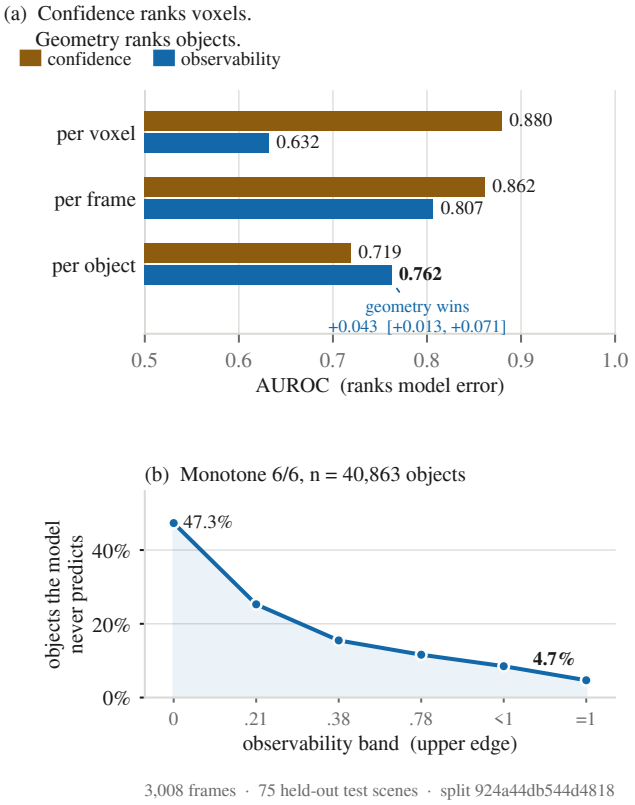


Figure 3. **Confidence ranks voxels; geometry ranks objects.** (a) AUROC for ranking model error at three granularities on the sealed test split. The model's own confidence dominates geometry per voxel and per frame, then *loses* per object. (b) The mechanism: the fraction of annotated dynamic objects the model never predicts, against observability, monotone across all six bands over 40,863 objects. Confidence cannot flag what the model never predicted.

2.2. Uncertainty and failure prediction

Maximum softmax probability [23] is still the baseline to beat on in-distribution misclassification, and recent work finds it competitive with much later proposals [63]. Temperature scaling [20] corrects calibration without touching the ordering, so it cannot change a ranking result. Ensembles and MC dropout [16, 34] estimate epistemic uncertainty at a multiple of inference cost; the aleatoric-epistemic split has been examined for vision [31]. A second family scores a sample post hoc: feature-space distance [35], neighbourhood agreement [29], input perturbation [42], free energy [44], and soft-probability tests [19]. A third trains the failure predictor itself [10, 18]. Selective prediction [17] gives the evaluation we adopt and conformal prediction [1] the distribution-free alternative. Dense prediction has its own reliability literature with the same shape [3, 12, 13].

Every method above reads the network. Each is a statement about a prediction the network produced, which is the limitation Fig. 2 describes and Tab. 2 measures.

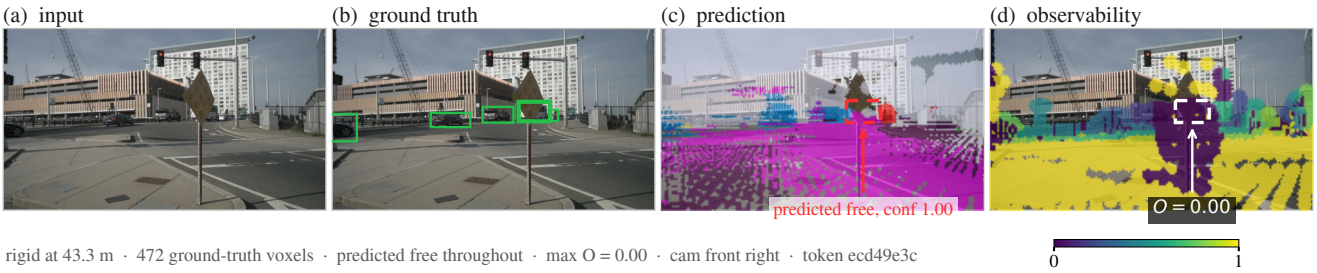


Figure 4. **A whole bus, predicted as free space, with maximal confidence.** One sealed-test frame, token `ecd49e3c`, in image space. (a) The camera the object falls in. (b) Occ3D ground truth, dynamic boxes projected; the rigid bus at 43.3 m, thick box, occupies 472 annotated voxels. (c) FB-OCC's predicted occupancy over the same volume, nearest surface, coloured by class: no dynamic class is predicted anywhere inside that box and the model assigns *free* there at confidence 1.00. Softmax offers no signal, because a confident prediction of free is exactly what a missed object looks like. (d) Geometric observability sampled on the ground-truth surface. The marked region is the darkest in the frame, $O = 0.00$, while the open road either side of it saturates. The geometry knows the volume is unobserved before the network runs, and knows it without ever reading the network's output.

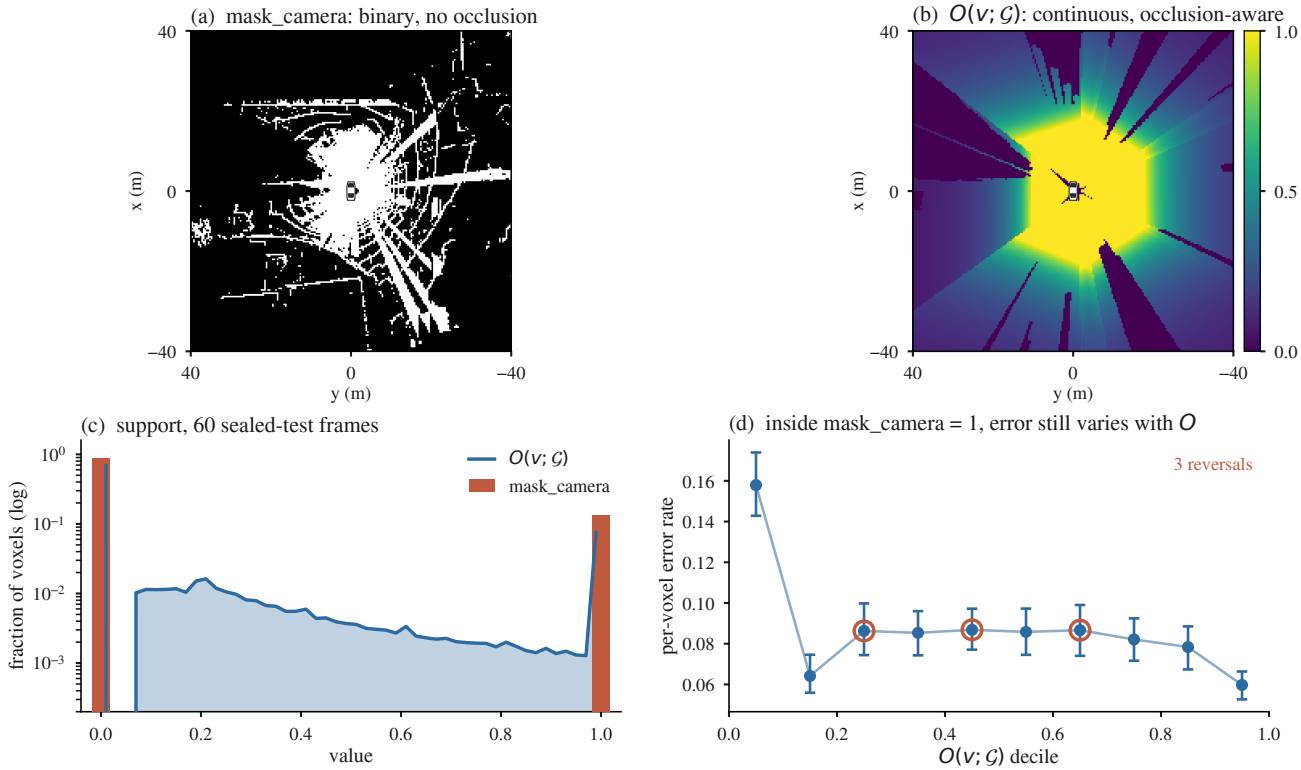


Figure 5. **`mask_camera` and $O(v;\mathcal{G})$ are not the same object.** (a, b) One sealed-test frame, token `4ae9c033`, bird's eye. The dataset mask is binary and is built offline from the full sensor suite, including the LiDAR the model never receives; observability is continuous and resolves occlusion by first-hit ray marching, so the shadow wedges behind buildings are explicit. (c) Over 60 sealed-test frames the mask takes two values and observability occupies the interval, log scale. (d) Restricting to voxels the mask already calls observed, the per-voxel error rate still varies with O by a factor of 2.6×: the mask is constant there and ranks nothing. The relation is not monotone (3 deciles rise, circled) and is consistent with the H1 contrast in Tab. 7, which scores the *calibration gap* against the same deciles rather than the error rate and counts its own reversals. The effect is concentrated in the lowest decile. Error bars are paired-frame bootstrap, 95%.

2.3. Visibility, and the camera mask

Visibility is not new here and we claim no priority over it. Occ3D [52] ships a camera-observation mask; EvOcc mod-

els unobserved and conflicting space evidentially [30]; α-OCC studies uncertainty for camera occupancy directly [49]; sensor visibility has been formalised with its own metrics [5]; and datasets exist to probe occlusion specifically [33]. Driving world models [48, 51, 56, 64] treat geometry as something to reconstruct or condition on. In all of these, visibility is an input to a model or a mask on a loss.

The camera mask is the closest comparison, and it is one bit. Occ3D's `mask_camera` marks whether a voxel is observed in the current camera view, and the obvious objection to this paper is that it already encodes what we measure. Three differences decide the matter. It is binary, so inside the mask it is constant and ranks nothing, which is the regime our comparison operates in. It is built offline from the full sensor suite, including the LiDAR the model never receives, so it describes the annotation pipeline rather than what the cameras resolved. And it marks observation rather than evidence strength: a voxel grazed by one camera at 40 m and a voxel filling half an image both read 1. Figure 5 shows the consequence. Restricted to voxels the mask already calls observed, the per-voxel error rate still varies with observability by a factor of 2.6×. None of this means the mask fails at its own job, which is deciding where labels are valid.

What separates this work is the role the quantity plays: held outside the model and evaluated against the model's own uncertainty, at three granularities, with the geometry source stated and both choices reported.

3. Geometric Observability

A camera rig either resolves a volume of space or it does not, and that fact is settled by the calibration and the scene, not by the network. This section makes it a number. We first fix the problem setting (Sec. 3.1), then define the measure (Sec. 3.2), then state what it costs to compute (Sec. 3.3), and finally separate the two sources the scene geometry can come from (Sec. 3.4), because that choice decides what the measure can be used for.

3.1. Problem setting

A camera-only occupancy model takes the images $\{I_i\}_{i\in\mathcal{C}}$ from a calibrated rig of $|\mathcal{C}| = 6$ cameras and predicts a class for every voxel of a fixed grid $\mathcal{V}$ around the ego vehicle. On Occ3D-nuScenes the grid is $200 \times 200 \times 16$ at 0.4 m, covering ± 40 m laterally, with eighteen classes of which one, index 17, means `free`. Write the prediction $\hat{y}(v)$ and the ground truth $y(v)$.

We study two failure events. A *voxel error* is the event $\hat{y}(v) \neq y(v)$, that the predicted class of a voxel differs from its ground-truth class. A *missed object*, written M_j and defined in Eq. (1), is the stronger event that a whole annotated instance goes unrepresented: for a ground-truth box $\mathcal{B}_j$ of a dynamic class,

$$M_j \;=\; \mathbb{1}[\,\nexists\, v \in \mathcal{B}_j \;:\; \hat{y}(v) \in \mathcal{D}\,], \tag{1}$$

where $\mathcal{D}$ is the set of dynamic classes. The two events are not the same question and, as Sec. 4.2 shows, they are not ranked by the same signal.

A failure-ranking signal is any score $\phi(\cdot)$ computed per voxel or per object, evaluated by how well it separates failures from non-failures. The incumbent is the model's own maximum softmax probability [23], $\phi_{\text{MSP}}(v) = \max_c \hat{p}_c(v)$, negated so that low confidence predicts error. Its structural limitation is visible in Eq. (1): when an object is missed, every voxel inside its box carries a confident vote for `free`, so ϕ_{MSP} is not merely uninformative there, it points the wrong way. Any signal that is to catch this event cannot be a function of $\hat{p}$.

3.2. The measure

Two quantities enter the measure. The *projected coverage* $c_i(v) \in [0, 1]$, defined in Eq. (3), is how much image area camera i devotes to voxel v, and so how well that camera resolves it. The *first-hit visibility* $s_i(v; \mathcal{G}) \in \{0, 1\}$, defined in Eq. (4), is whether the ray from camera i reaches v before anything else in the scene-geometry field $\mathcal{G}$. Geometric observability $O(v; \mathcal{G})$ is then the best evidence any one camera supplies about v:

$$O(v; \mathcal{G}) \;=\; \max_{i\in\mathcal{C}} \; c_i(v) \cdot s_i(v; \mathcal{G}). \tag{2}$$

The maximum rather than a sum encodes that a voxel seen well by one camera is observed, however many others miss it. Nothing in Eq. (2) reads $\hat{p}$, or any uncertainty head, or any learned parameter.

Coverage. A voxel of side r at depth d along the optical axis of a camera of focal length f projects to a footprint of roughly $(fr/d)^2$ pixels. Coverage normalises that by a constant and clips:

$$c_i(v) \;=\; \min\!\left(1,\; \frac{(f_i r/d_i(v))^2}{\pi_{\text{full}}}\right), \qquad \pi_{\text{full}} = 965.0. \tag{3}$$

A voxel resolved by at least π_{full} pixels counts as fully covered; beyond that range evidence degrades as $1/d^2$ rather than falling off a threshold. This is the measure's only parameter. It was fixed once before any result in this paper and never retuned per backbone or per split, and Sec. 4.6 varies it over a factor of four.

First-hit visibility. Visibility is decided by marching the ray from camera i and asking whether v is the first occupied cell it meets:

$$s_i(v; \mathcal{G}) \;=\; \mathbb{1}\!\left[\, v = \arg\min_{u\in\mathcal{R}_i(v),\, \mathcal{G}(u)=1} \|u - o_i\| \,\right], \tag{4}$$

with $\mathcal{R}_i(v)$ the voxels the ray traverses and o_i the camera centre. The predicate $\mathcal{G}(u) = 1$ is the whole dependence on scene geometry, and Sec. 3.4 is about where it comes from.

Why not a z-buffer. Rendering the volume once per camera and reading depths would be far cheaper, and we tried nine variants of it. All of them decide occlusion once per voxel, and a voxel can be half-occluded: a pole in front of a bus removes part of the bus's footprint, not all of it. Resolving that correctly requires per-pixel first-hit, which is why Eq. (4) is evaluated by ray marching.

3.3. Computing it

The grid, the intrinsics and the extrinsics are identical in every frame; only $\mathcal{G}$ changes. The traversal order of each ray can therefore be precomputed once per camera and reused. We build that table in 6.9 s, hold it in 831.6 MB, and per frame the marcher becomes a gather and a prefix test over six cameras. Measured over eight frames this runs at 0.329 s per frame against 1.139 s for the direct implementation, a speed-up of 3.46×, and builds the full 6,019-frame pack in 33 rather than 114 minutes, single process on CPU. The two implementations were compared voxel by voxel over 30,720,000 voxels and disagreed on none.

Cost is $O(|\mathcal{C}| \cdot |\mathcal{V}| \cdot L)$ for an average ray length L, with no dependence on the occupancy model, so the same table serves any backbone evaluated on the same rig and grid.

3.4. Where the scene geometry comes from

Equation (4) needs to know what else is solid, and that requirement is not a detail. We evaluate two choices and keep them notationally distinct throughout, as $O_{\text{GT}} = O(v; \mathcal{G}_{\text{GT}})$ and $O_{\text{pred}} = O(v; \hat{\mathcal{G}}_{\text{pred}})$.

$\mathcal{G}_{\text{GT}}$ is the dense Occ3D ground-truth volume. It is independent of the occupancy model, which is why the resulting measure transfers unchanged between architectures, but it is not available on a vehicle. It is an evaluation-time quantity and every number in Sec. 4.2 uses it.

$\hat{\mathcal{G}}_{\text{pred}}$ is the model's own predicted occupancy, thresholded to occupied or free. This one *is* available at inference. Note that even here observability reads only the hard class decision, never $\hat{p}$, so the comparison in Sec. 4.2 stays a comparison between two kinds of information rather than two readings of the same one. Section 4.4 measures what happens under the substitution, and the answer is negative in a specific and informative way.

3.5. From voxels to objects

The missed-object event of Eq. (1) is defined on a box, so the signal has to be defined on a box too. For each annotated dynamic instance we take the voxels whose centres fall inside the oriented box, $\mathcal{B}_j$, and aggregate:

$$O_j = \max_{v \in \mathcal{B}_j \,:\, y(v) \in \mathcal{D}} O(v; \mathcal{G}), \qquad \phi_j^{\text{MSP}} = \max_{v \in \mathcal{B}_j} \max_{c} \hat{p}_c(v). \tag{5}$$

Both use a maximum, for the same reason and to keep the comparison fair: one well-resolved voxel is enough to say the rig had evidence, and one confident voxel is enough to say the model committed. Restricting O_j to voxels the ground truth assigns to the object keeps the score on the object rather than on the free space around it; the confidence score is not restricted, because the model does not know which voxels those are. Instances with fewer than eight ground-truth voxels are dropped, since below that the aggregation is dominated by discretisation. This leaves 40,863 objects on the sealed test split.

We also report a LiDAR point-count oracle, the number of returns inside $\mathcal{B}_j$. It sees a modality the model never receives and is an upper reference rather than a competing method.

4. Experiments

The experiments answer four questions in order. Does the measure beat the dataset's own camera mask at ranking error (Sec. 4.2)? Does its value depend on the granularity of the failure being ranked (Sec. 4.3)? Is it redundant with the model's confidence, or a second axis? And does it survive the only occupancy field a vehicle actually has (Sec. 4.4)? Every number is computed on a split sealed before method development, and every contrast was specified before its data existed.

4.1. Setup

Models and data. Occ3D-nuScenes [6, 52] supplies a 200 × 200 × 16 grid at 0.4 m around the ego vehicle, 18 classes and six cameras. FB-OCC r50 [41] with 16-frame temporal fusion is the primary model; we measure mIoU 38.90 against a published 39.1, so the checkpoint reproduces. FlashOcc [57] R50 in its 1-frame (M1) and 2-frame (M2) configurations is a second architecture, used to test whether the voxel-level comparison replicates. All models use publicly available pretrained checkpoints, and the observability maps are computed once and shared across all three.

The sealed split. The 150 validation scenes are partitioned once into a development half and a held-out test half of 75 scenes, recorded under digest `924a44db544d4818`. Method development used the development half only; the test half was opened once. A single module is the source of truth for the split, and every script and every CI gate reads it from there.

Statistics. AUROC [21] by exact histogram on quantised scores; the paired scene bootstrap [14] for every interval,

Model (frames)	SIGHTOCC	mask	[95% CI]	in mask
FB-OCC (16)	0.6714	0.6230	[+.039, +.058]	0.6321
FlashOcc M2 (2)	**0.7041**	0.6340	[+.060, +.079]	0.6414
FlashOcc M1 (1)	0.6908	0.6299	[+.052, +.070]	0.6365

Table 1. **Per-voxel error ranking, test split, three camera-only models.** AUROC of observability and of the binary `mask_camera` over the full volume, with the paired scene-bootstrap interval on their difference. The last column scores observability *inside* the mask, where the mask is constant and ranks nothing. 3,008 frames, 75 scenes.

which resamples whole scenes rather than voxels because voxels within a scene are not independent; risk-coverage and AURC for selective prediction [15, 17]; ECE in 15 bins [45]. Every contrast reported in this paper was specified before the corresponding data existed, in the sense of [46].

4.2. Main results

Observability outranks the camera mask. Table 1 scores per-voxel error over the full volume, using observability and the dataset's binary mask each as a ranking score. SIGHTOCC outranks the mask on all three models, by $+0.0484$ to $+0.0700$ AUROC, every interval excluding zero. The acceptance bar for the second architecture, an interval excluding zero with a point estimate above $+0.02$, was fixed before any FlashOcc code was run. The margin is not monotone in temporal depth across 1, 2 and 16 frames, so it is not a property of temporal fusion, and we do not claim the margins differ between models.

Inside the mask, where the mask is constant and therefore ranks nothing, observability still reaches 0.6321 to 0.6414. The headline contrast H2 is +0.0484 [+0.0390, +0.0584] on the test half, and $+0.0534$ on the full 150 scenes; the test figure is the one we quote, and the two are given together so the difference cannot be mistaken for a selection effect.

This is a modest effect in absolute terms and we do not inflate it. Its interest is that it is obtained by a quantity that costs no training, reads nothing from the network, and is available before the forward pass.

4.3. The granularity inversion

What counts as a miss. For a ground-truth box B_j of class y_j with predicted labels $\hat{y}$:

$$M_j^{\text{any}} = \mathbb{1}[\,\nexists\, v \in B_j : \hat{y}(v) \neq \textit{free}\,], \tag{6}$$

$$M_j^{\text{dyn}} = \mathbb{1}[\,\nexists\, v \in B_j : \hat{y}(v) \in \mathcal{D}\,], \tag{7}$$

with $\mathcal{D}$ the dynamic classes. We call M^{any} the **strict miss**, nothing at all predicted inside the box, and M^{dyn} the **dynamic miss**, nothing *movable* predicted there, so a box filled with predicted road surface counts as a miss. The second is the one a planner cares about and the terms are used consistently below. We report M^{dyn} and give M^{any} beside it, because the choice changes the rate and a reader should see both. Boxes with no LiDAR return are excluded, and only boxes holding at least one ground-truth voxel of their own class are scored.

Figure 4 is one such object in full: the ground truth contains a bus, the prediction contains no dynamic class anywhere inside its box, and the observability field is near zero over the same cells. The strict and dynamic criteria agree on that example; they disagree on boxes the model fills with static classes, which is why both are reported.

Two rates, two conditionings. Across all dynamic objects in the test split, 19.3% are missed under M^{dyn}. *Conditional on the object's voxels having zero observability* it is 47.3%, and at full observability 4.7% (Fig. 3b). The figures are not in tension and we state the conditioning each time.

At the voxel level, maximum softmax probability reaches 0.8797 against observability's 0.6321. At the frame level the ordering holds but narrows, 0.8618 against 0.8066. At the object level it reverses: observability reaches 0.7623 against confidence's 0.7193, a gain of +0.0432 [+0.0135, +0.0708] over 40,863 objects in 3,008 frames. Every paired contrast in this paper, this one included, is the mean of the paired scene-bootstrap difference distribution with its percentile interval, not the difference of the two point estimates printed beside it. Here those estimates differ by $+0.0431$ and the bootstrap mean is $+0.0432$; we quote the bootstrap mean throughout because it is the quantity the interval is built around.

A LiDAR point-count oracle reaches 0.7745. It counts returns inside the ground-truth box in the current sweep and sees a modality the model never receives, so it is an upper reference, not a competitor. The paired contrast is -0.0121 $[-0.0289, +0.0052]$: observability is *statistically indistinguishable* from it on this test set, which is weaker than matching it and is what the interval supports.

Table 2 breaks this down. Observability wins decisively on car, the largest class by a factor of two ($+0.152$), and on truck ($+0.089$). It loses on trailer (-0.198) and bicycle (-0.115), and ties on the rest. Both losses are consistent with the mechanism rather than against it: a trailer is large enough that some part of it is almost always observed, so observability is near-constant across the class and has little left to rank, while the model's confidence still varies.

Object-level baselines. A single confidence baseline is not enough to support the claim, so Tab. 3 scores every signal we can read from the same predictions on the same objects.

The model-derived family is not close to exhausted by maximum softmax, but it is bounded by it: the best of the three reaches 0.7193 and the other two sit near chance, because inside a box the model never populated there is very

Table 2. **Per class, object level, sealed test split.** *missed* is the fraction of annotated objects the model never predicts. AUROC columns rank that event. ORACLE counts LiDAR returns inside the box and therefore sees a modality the model does not; it is an upper reference, not a competitor. Observability wins on the two largest classes and loses on trailer and bicycle, and both losses are printed.

Class	n	missed	AUROC ↑			obs−conf
			conf.	obs.	ORAC.	Δ
car	22,096	11.8%	0.681	**0.832**	0.786	**+0.152**
pedestrian	10,407	36.2%	**0.711**	0.707	0.739	-0.004
truck	3,956	11.6%	0.696	**0.786**	0.765	**+0.089**
trailer	1,056	11.5%	**0.776**	0.553	0.630	-0.198
motorcycle	1,011	25.4%	0.673	**0.720**	0.714	+0.047
bicycle	982	33.2%	**0.820**	0.708	0.723	**-0.115**
bus	814	13.3%	**0.731**	0.723	0.744	-0.001
construction vehicle	541	48.2%	**0.729**	0.646	0.656	-0.087
all dynamic	40,863	19.3%	0.719	**0.762**	0.774	**+0.043**

	Signal	AUROC ↑
model	max softmax probability	0.7193
model	mean confidence in box	0.5409
model	mean top-1 minus top-2 margin	0.5202
geom.	range from ego	0.6660
geom.	observability O_{GT}	**0.7623**
	LiDAR points in box (ORACLE)	0.7745

Table 3. **Object-level failure signals, FB-OCC, test split.** Every signal is scored on the same 40,863 objects for ranking the missed-object event of Eq. (7); higher is better. The LiDAR oracle sees a modality the camera-only model never receives and is an upper reference, not a competitor.

little for any output statistic to vary over. Entropy and an energy score require the full class distribution, which our export does not retain, and MC-dropout and ensembles require additional forward passes, so all four remain untested here.

4.4. The geometry source

The result above uses occ_{gt} and is a statement about an evaluation-time measure. We rebuild observability with occ_{pred}, changing nothing else: same marcher, same π_{full}, same sealed split. The acceptance criterion was registered before the run: the paired interval on $\text{AUROC}(\text{obs}_{\text{pred}}) - \text{AUROC}(\text{MSP})$ at object level must exclude zero.

It does not survive. Table 4 reports the comparison on a pre-declared subsample of 800 test frames (seed fixed before the run), 10,790 objects in all 75 test scenes.

Paired scene bootstrap: occ_{gt} reproduces the headline on this subsample, $+0.0464$ $[+0.0193, +0.0733]$ over MSP, which confirms the subsample is not the cause of what follows. The deployable form is -0.0477 $[-0.0828, -0.0135]$ *below* MSP, and -0.0936 $[-0.1113, -0.0761]$ below its own evaluation-time counterpart. The deployable substitution therefore does not preserve the object-level advantage, and the next subsection isolates why.

Occluder for s_i	AUROC ↑
occ_{gt} (evaluation-time)	**0.7649**
MSP (no geometry at all)	0.7184
occ_{pred} (deployable)	0.6708

Table 4. **Which geometry resolves the occlusion.** Object-level AUROC on the 800-frame test subsample when the first-hit test of Eq. (4) is resolved against the ground-truth volume, against nothing at all, and against the model's own predicted occupancy.

mean observability	occ_{gt}	occ_{pred}	Δ
missed objects	0.1987	0.2017	$+0.0030$
detected objects	0.5402	0.4586	$\mathbf{-0.0816}$
separation	0.3415	0.2569	-25%

Table 5. **Which side of the comparison moves.** Mean observability over the 10,790 objects of the test subsample, resolved against each occluder. The substitution leaves missed objects where they were and pulls detected objects down, so the separation that carries the ranking shrinks by a quarter.

Which side of the comparison moves. The intuitive explanation is that a missed object, being predicted free, stops occluding, so the ray passes through and observability is falsely *high* exactly where we need it low. That explanation is wrong, and the per-object means say so (Fig. 6c and Tab. 5).

Observability at missed objects is unchanged. What collapses is observability at *detected* objects, and the ranking degrades because the top of the scale falls rather than because the bottom rises.

The cause is density. Predicted occupancy is far coarser than the ground-truth volume (over 60 sampled frames the occupied fraction is 0.579 against 0.052, a factor of 11, and $37\times$ on the frame in Fig. 6), so a great many more rays are stopped. Mean observability on that frame falls 41% and the zero-observability fraction rises from 48.9% to 77.7%. The measure saturates toward zero and loses the range it needs to discriminate. This is over-occlusion, not lost evidence.

That distinction matters for what comes next. An evidence-erasure failure would be circular and unfixable; an over-occlusion failure is a property of *this* occupancy field, and a sparser, better-calibrated source of geometry independent of the occupancy model is not ruled out by this result. Monocular metric-depth foundation models are the obvious candidate [4, 24, 54]: they run at inference, and they do not inherit the occupancy model's errors, so a pedestrian that model drops still has a surface in the depth map. A prior

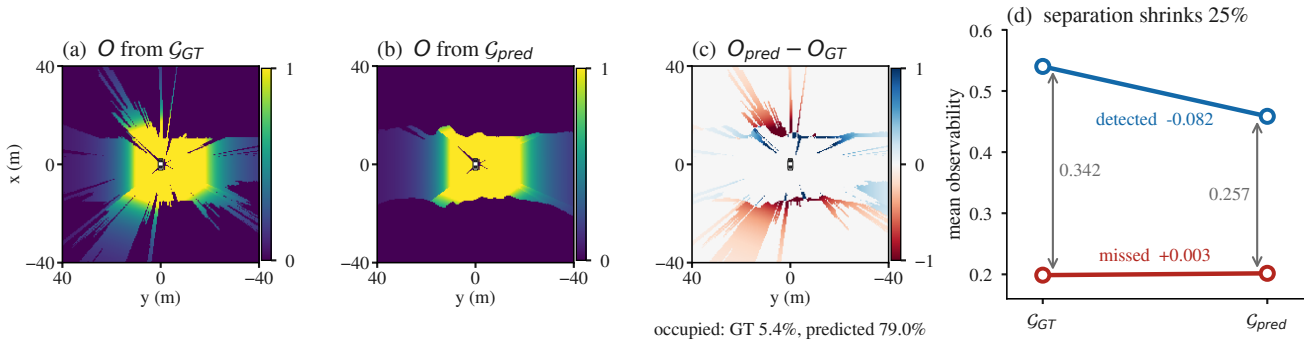


Figure 6. **What changes when the occluder is predicted.** (a) Observability resolved against the ground-truth volume, one sealed-test frame, token `90c191b4`. (b) The same frame resolved against the model's own predicted occupancy, same colour scale: that field calls 79.0% of voxels occupied against 5.4% in the ground truth, so it casts far more shadow. (c) Their difference. Red is observability *lost* to the substitution, and it dominates: this is over-occlusion, not the circularity one would predict. (d) The measured mechanism over 10,790 objects. Observability at *missed* objects is unchanged, while at *detected* objects it collapses, so the separation between the two shrinks by a quarter and the ranking degrades. Object-level AUROC falls from 0.765 with $\mathcal{G}_{GT}$ to 0.671 with $\mathcal{G}_{pred}$, below the 0.718 of plain softmax.

map or a second modality would serve the same role. We have tried none of them and claim nothing about them.

4.5. Analysis

The two signals are not redundant. If observability were merely a noisy restatement of confidence, holding one fixed would flatten the other. It does not (Fig. 7a). Among objects the model is *confident* about, those in the low-observability half are missed 23.6% of the time against 3.7% in the high-observability half, a factor of 6.3 at matched confidence. The reverse also holds, so neither axis subsumes the other.

One number in that figure deserves its own sentence: the median maximum-softmax value over all test objects is 1.00. More than half of every annotated object in the split sits at saturated confidence. A score that is maximal for the median case cannot separate the median case from anything, and that is the structural reason the object-level comparison goes the way it does.

A caveat against our own headline. The ground-truth volume is sparse partly because unobserved space is labelled free, and an occluder set that sparse may *under*-occlude, which would make $\mathrm{occ}_{\mathrm{gt}}$ optimistic. We have not quantified this and state it because it cuts against us.

The combined score. Where a combined score appears it is a logistic regression on four features, fitted on the development half and applied unchanged to the test half:

$$S(v) = \sigma\big(w^{\top}[\,\mathrm{logit}\,p_{\max}(v),\ m(v),\ \mathrm{obs}(v),\ d(v)\,] + b\big), \tag{8}$$

with $p_{\max}$ the maximum softmax probability, m the top-1 minus top-2 margin and d range from ego. No weight is tuned on the test half.

Score	AURC ↓
confidence alone	0.2526
confidence + `mask_camera`	**0.2128**
confidence + observability	0.2160
observability alone	0.3156
`mask_camera` alone	0.3542

Table 6. **Selective prediction on the sensor-only volume.** Area under the risk-coverage curve; lower is better. Observability is useful as an addition to confidence and poor as a replacement for it.

Downstream: selective prediction. A ranking score is only useful if abstaining on its worst-ranked fraction actually lowers risk. Measuring area under the risk-coverage curve (AURC, lower is better) on the sensor-only volume gives Tab. 6.

Adding observability to confidence reduces AURC by $0.0365\ [0.0307, 0.0425]$, which holds. Against confidence plus `mask_camera` it is $-0.0033\ [-0.0051, -0.0015]$: **observability loses this comparison, and the interval excludes zero.** We report it because an oracle mask built from the full sensor suite offline ought to win a volume-level abstention test, and because the object-level result in Tab. 2 is not weakened by it. The two measure different things, and only one of them is available on a vehicle.

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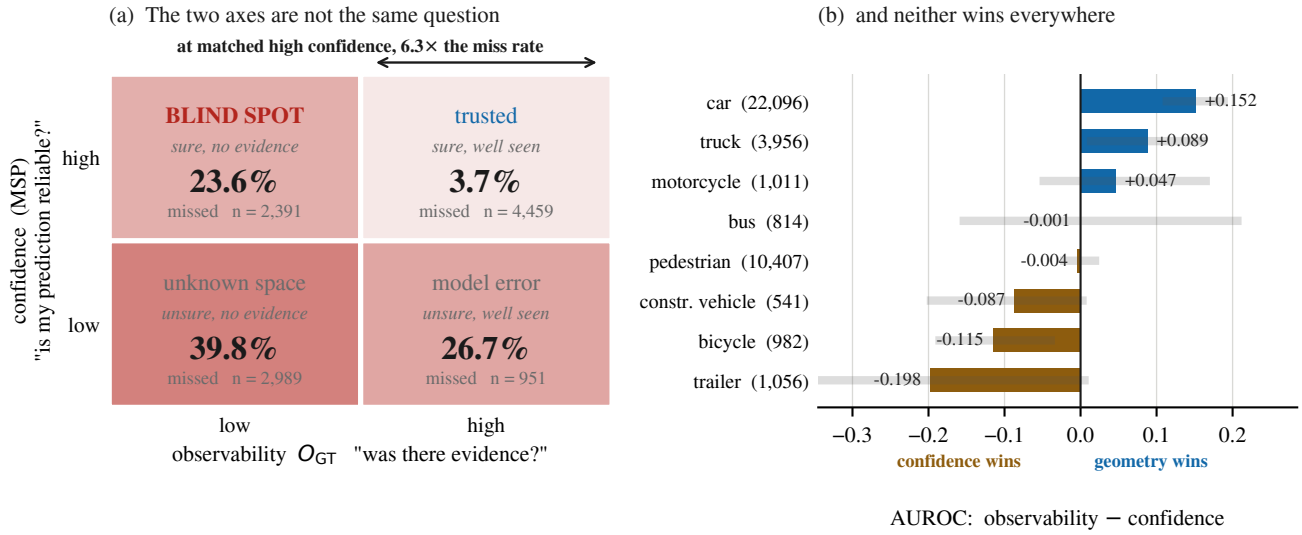


Figure 7. **Confidence and observability are different axes, populated with real objects.** (a) Test objects split at the population medians, O 0.36 and MSP 1.00 (more than half of all objects carry the maximum softmax value, which is the problem the paper starts from); shading tracks the miss rate. The taxonomy one would draw is not what the data gives: the worst cell is low confidence with low observability, not high confidence with low observability. What the data does support is the paper's claim. Holding confidence *high* and varying observability gives 23.6% against 3.7% missed, a 6.3× difference at matched confidence, so the two axes are not redundant. (b) Per class; bars are the paired scene-bootstrap mean and rules the 95% interval. Observability wins decisively on car and truck, ties on pedestrian and bus, and loses on construction vehicle, bicycle and trailer. Neither signal dominates, and we report the losses at the same size as the wins.

Table 7. **All measured contrasts.** Paired scene-bootstrap means with 95% intervals on the sealed test split (75 scenes, 3,008 frames); AUROC difference unless noted. Each contrast was specified before its data existed, and every one that was specified is listed here.

	Contrast	ΔAUROC [95% CI]
Per voxel		
H2	Observability vs. `mask_camera`	+0.0484 [+0.0390, +0.0584]
H4	Temporal vs. instantaneous observability	+0.0156 [+0.0098, +0.0214]
H5	Combined score vs. observability alone	+0.2491 [+0.2407, +0.2575]
H6	Combined score vs. model confidence	+0.0015 [+0.0009, +0.0022]
H1	Calibration gap vs. observability decile	$\rho = -0.433$, 2 reversals
Per object		
E1	Observability vs. model confidence	+0.0432 [+0.0135, +0.0708]
E2	Observability vs. LiDAR point-count oracle	-0.0121 [-0.0289, +0.0052]
A55	Predicted-geometry occluder vs. model confidence	-0.0477 [-0.0828, -0.0135]
Selective prediction and calibration		
S1	Observability vs. confidence, selective risk	+0.0365 [+0.0307, +0.0425]
S2	Observability vs. `mask_camera`, selective risk	-0.0033 [-0.0051, -0.0015]
R1	Observability-conditioned vs. global recalibration, ECE	+0.0026 [-0.0026, +0.0058]

4.6. Ablations

Temporal accumulation. Propagating past observability through ego poses with an exponential half-life, and scoring on the same voxels: half-lives of 0.5, 1, 2 and 4 s give +0.0080, +0.0083, +0.0156 and +0.0196 AUROC over the instantaneous map. The half-life was fixed at 2 s before the run; at that value the interval is [+0.0098, +0.0214] and H4 holds. The monotone trend to 4 s is reported but was not registered and is not claimed.

Does π_{full} matter? Less than one might fear, and the analytic part bounds the empirical part. Coverage is $\min(1, A_i/\pi_{\text{full}})$, so for a voxel where no camera saturates, changing π_{full} multiplies every c_i by one constant, leaves the $\arg\max$ camera unchanged, and is a strictly monotone transform of the score. Every rank metric here is *exactly* invariant on that set, and π_{full} can act only through the saturated remainder, which is 11.9% of in-frustum voxel–camera pairs at the registered value. Table 8 rebuilds observability end to end over a fourfold range and scores it at object level

π_{full}	object AUROC ↑	Δ
500	0.7485	−0.0045
965 (registered)	0.7529	(ref.)
2000	0.7552	+0.0022
MSP, same objects	0.7366	

Table 8. **Sensitivity to the one parameter.** Object-level AUROC with π_{full} varied over a factor of four, rebuilt end to end. Δ is measured against the registered value, which is the reference row. Every setting beats maximum softmax on the same objects.

Variant	AUROC ↑	vs `mask_camera`
$\max_i c_i s_i$ (ours)	**0.6755**	+0.0575
sum over cameras	0.6748	+0.0568
no temporal term	0.6755	+0.0575
no visibility term s_i	0.5908	**−0.0272**
`mask_camera`	0.6180	(ref.)

Table 9. **Which term carries the result.** Per-voxel AUROC for each variant of Eq. (2), compared against the dataset's binary mask, which is the reference row. Removing the visibility factor s_i is the only change that matters, and it drops the measure below the mask.

on 2080 objects.

The spread across 4× is 0.0067 AUROC, smaller than the effect being reported, and every value beats MSP. The registered 965 was derived to preserve a 16.3 m saturation radius before any of these numbers existed, and no other value is used anywhere in the paper.

Temperature scaling is not a competitor. It is a monotone transform of the confidence score, so it leaves every ranking metric in this paper exactly unchanged. It therefore appears in no ranking table in this paper.

Which term carries the result. Equation (2) has two factors and a pooling choice, and only one of the three matters. Table 9 scores per-voxel error on 675 frames across 145 scenes.

Pooling by maximum rather than by sum is worth +0.0007, which is nothing; we keep the maximum because it is the one with a reading ("seen well by at least one camera"), not because it measures better. Dropping the visibility factor, so that coverage alone decides, costs 0.0847 AUROC and drops the score *below* `mask_camera`. Resolving occlusion is not a refinement of this measure. It is the measure.

What the blind volume is made of. On 1,238 frames, 76.6% of ground-truth obstacle voxels fall in volumes with no camera evidence, which is large enough to deserve an account. Attributing each shadowed voxel to the surface that casts it, over 400 frames and 169M occluded voxels: man-made structure 43.9%, vegetation 27.9%, other vehicles 10.4%, drivable surface 4.3%, remainder spread across the rest. The blind volume is mostly buildings and foliage, with one voxel in ten hidden behind another car, which is the subset a planner cares about most.

Would a seventh camera fix it? We re-ran the geometry with each of six additional camera placements added to the rig. The best of them, a higher roof mount, recovers 2.23% of those voxels; a rear telephoto recovers 1.29%; no other placement exceeds 0.8%. A voxel counts as recovered when its observability rises from exactly zero to any positive value. Six placements are a finite search and license no universal claim; what we can say is that none of the placements evaluated closes more than a small fraction of the unobserved volume.

Summary of measured contrasts. Table 7 collects every contrast in the paper with its paired bootstrap interval. Three are negative and are reported with the same weight as the positive ones: the calibration gap is not monotone in observability (H1), the mask is marginally better at selective risk than observability alone (S2), and the predicted-geometry occluder does not preserve the object-level advantage (A55). The first two are small enough that they bear on presentation rather than on the claim; the third defines where the measure applies, which is why Sec. 4.4 measures its mechanism rather than reporting it in passing.

5. Conclusion

Geometric observability, computed from the camera rig and an external occupancy field with one parameter and no learned component, reveals where a camera-only occupancy model fails in a way its own confidence cannot. Its value is not uniform across granularity, and that non-uniformity is the result: the model's own confidence is the better signal for whether a voxel is wrong, and the worse signal for whether an object is missing entirely, because a missed object is a confident prediction of free space. External geometry has no such blind spot, because it never consults the model. Rebuilding the same signal *from* the model's own predicted occupancy behaves differently, and not for the reason one would guess: observability at missed objects barely moves, while observability at detected objects collapses, because predicted occupancy is an order of magnitude denser than the ground-truth volume and shadows regions the rig genuinely sees. That is a property of the particular occupancy field rather than a circularity in the idea.

Limitations. (1) Observability is resolved against an external occupancy field. Substituting the model's own predicted occupancy makes it score below maximum softmax at object

level, so in the form evaluated here it is an evaluation-time diagnostic rather than a runtime monitor. Supplying that field from a source independent of the occupancy model, such as a monocular depth foundation model [4, 24], is the obvious next step and we have not tested one. (2) The object-level results are one checkpoint of FB-OCC r50 on Occ3D-nuScenes; only the voxel-level comparison is replicated on a second architecture, so Occ3D's annotation geometry defines both the ground truth and the occluder set. (3) Test scenes were used during exploration before the partition was enforced in code, so validation on a second dataset is still needed. (4) Visibility is binary per voxel–camera pair and object scores use a maximum; fractional visibility and other aggregations are untested. (5) We measure ranking quality, selective risk and one gating ablation, not collision rate, and we do not compare against sampling-based uncertainty baselines, which need forward passes our export does not retain.

Code and data. The evaluation harness, the sealed split and its digest, and the dated amendment log that fixes every hypothesis and acceptance threshold before its run are available upon request.

Use of large language models. A large language model was used in two ways. During the research it was used to discuss candidate experiments and to help interpret results. During revision of this manuscript it was used to edit and condense the text, to check reported numbers and method descriptions against the evaluation code and the result files, and to verify bibliographic entries against their published sources. The authors conceptualised the study, developed the methodology, designed the experiments, defined the acceptance criteria, ran the experiments, reviewed all edits and checks, and take full responsibility for the content.

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